



Glenohumeral arthrodesis outcomes and complications: a systematic review

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Background: The purpose of this study was to systematically review the evidence in the literature to determine the clinical outcomes following glenohumeral arthrodesis.

Methods: Two independent reviewers performed a literature search in the PubMed database based on Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines. Studies were included if they reported on outcomes following shoulder arthrodesis from the years of 2000–2022. Functional outcomes that were collected included the Subjective Shoulder Value, American Shoulder and Elbow Surgeons shoulder index score, visual analog scale pain score, and Oxford Shoulder Score. Range-of-motion data were also collected.

Results: This review included 17 studies, with a total of 316 patients, that met the inclusion criteria. The majority of the patients were male (67.4%), and the average age was 38.4 years (range, 7–82 years). The overall fusion rate was 88.7%, and the time to fusion was on average, 3.9 months (range, 2–8 months). Shoulder arthrodesis resulted in improvement in each of the functional outcomes assessed: Subjective Shoulder Value (preoperatively, 18.8; postoperatively, 43.9; and percent change, 132.8%), American Shoulder and Elbow Surgeons shoulder index score (postoperatively, 62.1), visual analog scale pain score (preoperatively, 8.5; postoperatively, 3.03; and percent change, 62.4%), and Oxford Shoulder Score (preoperatively, 9.4; postoperatively, 30.9; and percent change, 328.7%). The abduction, forward flexion, external rotation, and internal rotation range-of-motion measurements postoperatively were 57°, 64°, 3°, and 48°, respectively. Complications were reported in 33.6% of patients, with fractures (20.9%) and infections (18.6%) being the most common sources of complication.

Conclusion: Shoulder arthrodesis provides improvement in functional outcomes for end-stage glenohumeral injuries; however, it is also associated with high rates of complications.

Level of evidence: Level IV; Systematic Review

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Shoulder arthrodesis is an end-stage salvage procedure for the glenohumeral joint and provides a stable shoulder

with varying functional outcomes.^{1,6} Owing to improved arthroplasty outcomes, including increased use of the reverse prosthesis, the use of shoulder arthrodesis has become less common in recent years.^{6,21} Common indications for shoulder arthrodesis include failed shoulder arthroplasty, chronic infection, reconstruction for refractory instability, and brachial plexus injuries, as well as other neurologic injuries such as stroke.^{1,21} Other candidates for

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shoulder arthrodesis include patients with debilitating pain, as arthrodesis aims to result in a pain-free shoulder that allows patients to continue to perform basic daily activities.^{7,21}

Fusion can be obtained via screws, plates, or external fixation with the goal of patients being able to complete activities of daily living after fusion, including using the hand to reach the mouth, to reach past the midline, and the reach the back pocket.¹ Despite the indications and goals, shoulder arthrodesis is not widely performed as revision shoulder arthroplasty has similar pain relief rates and improved functional outcomes.¹ Arthrodesis may still be considered in patients with many failed revision arthroplasties or glenohumeral arthritis with concomitant rotator cuff and deltoid muscle dysfunction to provide a more predictable outcome.²⁶ A significant downside to this procedure is that arthrodesis can result in a high rate of complications, including nonunion, malunion, infection, and fracture, as well as the need for subsequent surgical procedures.^{21,26} In salvage cases, there is growing discussion regarding the utility of arthrodesis after failed arthroplasties.^{1,21} Although the utility of arthrodesis needs to be clarified, new techniques, including arthroscopically assisted arthrodesis, are being developed, with similar outcomes to open arthrodesis.^{18,19} The development of these techniques means the utility of arthrodesis must be clarified as there is active innovation in the field.

The purpose of this study was to systematically review the evidence in the literature to determine the clinical outcomes following shoulder arthrodesis. The hypothesis of this study was that shoulder arthrodesis would result in acceptable functional outcomes but high complication rates.

Methods

Study selection

The literature search was conducted by 2 authors (K.A.-K. and A.Y.A.) using the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.¹³ The search results were reviewed, with a senior author (E.T.H.) arbitrating any variances. Titles and abstracts were screened, and potentially eligible studies underwent a full-text review.

Search strategy

The following search terms were used in the PubMed database in December 2022 as the search algorithm: (Shoulder) AND (Arthrodesis OR Fusion). A time limit from 2000-2022 was used as a part of the filtering criteria.

Eligibility criteria

The following inclusion criteria were used: (1) clinical study regarding shoulder arthrodesis, (2) clinical evidence on at least 5

patients, (3) publication in a peer-reviewed journal, and (4) publication in English. The exclusion criteria were (1) review studies, (2) cadaveric studies, (3) biomechanical studies, and (4) abstract-only publications.

Data extraction and analysis

Relevant information regarding study characteristics, including design and level of evidence, patient demographic characteristics, functional outcomes, and complications was collected by 2 blinded reviewers using a predetermined data sheet in Microsoft Excel (Redmond, WA, USA). The results were then tabulated to acquire the final data points. The level of evidence of each study analyzed either was directly taken from the article or was based on guidelines used by the Oxford Centre for Evidence-Based Medicine. Functional outcomes were selected based on commonly used objective outcomes to score pain relief and shoulder function. The outcomes measured included range of motion (ROM), visual analog scale (VAS) pain score, time to fusion, Subjective Shoulder Value (SSV), American Shoulder and Elbow Surgeons (ASES) shoulder index score, and Oxford Shoulder Score (OSS). The VAS pain score serves to quantify pain. Patients are asked to rate their pain on a scale of 0-10, with 0 being no pain and 10 representing the worst pain of their lives.¹¹ The SSV is used to assess patients' shoulder function and pain, with 0 being the worst outcome and 100 being the best outcome.¹⁰ The ASES shoulder index is a questionnaire that serves to standardize outcomes of different shoulder treatments by asking questions to both the provider and the patient.¹⁴ The OSS is a questionnaire that allows patients to rate their quality of life and function secondary to shoulder issues; the maximum score of the scale is 48, with higher numbers representing better outcomes.⁸ After data extraction, the aforementioned clinical outcome values were summated and averaged to yield the final output reported.

Results

Literature search

The initial literature search resulted in 1106 total studies. After a search to identify duplicates, the articles were screened based on the inclusion and exclusion criteria, yielding 17 unique studies and 316 unique patients. These were evaluated, and full texts were assessed for quantitative data. This process is further detailed in [Figure 1](#).

Patient demographic characteristics

There were 17 included studies with 316 patients.^{2-5, 8,9,12,16-20,22-26} All of the studies had either Level III or Level IV evidence. The majority of the 316 patients were male (67.4%), and the average age was 38.4 years (range, 7-82 years), with a mean follow-up period of 65.6 months (range, 3-432 months). The time to arthrodesis from injury was 4.0 months (range, 1.5-8 months), reported across 5 studies. The study characteristics and patient demographic characteristics are available in [Table I](#).

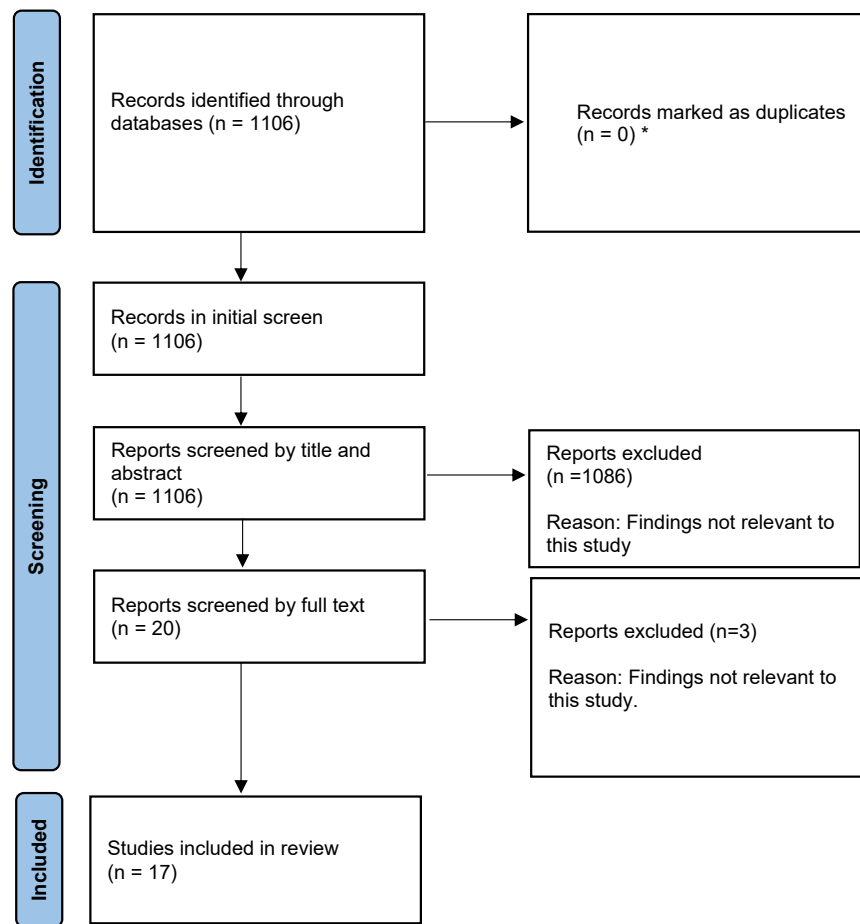


Figure 1 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) flow diagram of study inclusion and exclusion. *Covidence software (Melbourne, VIC, Australia) was used to verify the absence of duplicates after manual identification of no duplicates; however, it was not used for the remainder of screening.

Range of motion

No study reported on preoperative ROM. Postoperative ROM was reported in 10 studies (Table II). The ROM for abduction averaged 57° across 10 studies. The ROM for forward flexion was reported in 9 studies and was found at an average of 64° . The ROM for external rotation was recorded in 4 studies and was an average of 3° . The ROM for internal rotation was 48° across 6 studies.

Fusion

Overall, fusion rate and time to fusion were reported in 5 studies (Table III). The average fusion rate was 88.7% among 82 patients, and the average time to fusion was 3.9 months (range, 2-8 months).

Clinical outcomes

The SSV was reported in 4 studies. Preoperatively, the average SSV was 18.8 (n = 43); postoperatively, the SSV

increased to 43.9 (n = 51) (Table IV). The preoperative VAS pain score was collected in 1 study, with an average of 8.5 (n = 8), whereas 2 studies collected data on postoperative scores, showing a decrease in this value to an average of 3.0 (n = 26) (Table IV). No preoperative ASES scores were collected in the studies. ASES scores were collected after shoulder arthrodesis in 2 studies, with an average ASES score of 62.1 (n = 26) (Table IV). The OSS was recorded in 3 studies preoperatively and postoperatively (Table IV). The average OSS was 9.4 preoperatively (n = 15); postoperatively, this score improved to 30.9 (n = 33).

Complications and further surgical procedures

Further surgical procedures were reported in 8 studies; among these 8 studies (n = 191), 73 patients (38.2%) required at least 1 surgical procedure after arthrodesis (Table V). Complications were reported in 11 studies (n = 256), and 86 patients reported complications (33.6%) (Table V). Humeral fractures (n = 18, 20.9%) and

Table I Study and demographic characteristics

	Patients, N (n after exclusion of patients)	LOE	Male sex	Age, yr	Follow-up, mo	Previous surgery	Time to fusion, mo
Authors							
Atlan et al, ² 2012	57 (54)	III	57	24 (18-51)	42 (3-81)	0	NR
Belkheyar et al, ³ 2019	8 (8)	IV	5	28 (16-55)	20 (16-28)	6	4.5 (3-6)
Chammas et al, ⁴ 2004	29 (27)	IV*	25	25 (17-37)	71.2 (6-188)	NR	NR
Chib et al, ⁵ 2021	15 (15)	IV*	12	47 (42-55)	24 (18-NR)	NR	5.4 (4.3-7.1)
Diaz et al, ⁸ 2003	8 (8)	IV*	3	26 (18-36)	35 (24-83)	7	3.5 (2.5-5)
Dimmen and Madsen, ⁹ 2007	25 (18)	IV*	5	64 (19-75)	90 (36-180)	NR	NR
Lenoir et al, ¹² 2017	9 (8)	IV	9	33 (26-58)	28 (12-40)	8	3 (2-8)
Mimata et al, ¹⁶ 2015	5 (5)	III	4	21 (7-46)	36 (29-130)	NR	NR
Nagy et al, ¹⁷ 2004	27 (20)	IV*	12	47.1 (24.3-75.1)	63 (6-180)	12	NR
Porcellini et al, ¹⁸ 2014	12 (12)	IV	4	35 (21-49)	42 (24-60)	NR	NR
Puskas et al, ¹⁹ 2017	34 (34)	IV	19	50 (16-85)	43 (11-152)	26	NR
Rühmann et al, ²⁰ 2005	43 (43)	IV	33	35 (11-82)	80.4 (6-432)	5	3.5 (1.5-7)
Scalise and Iannotti, ²² 2008	8 (7)	IV*	3	47 (20-65)	55 (18-96)	4	NR
Sousa et al, ²³ 2011	19 (13)	IV*	11	46 (35-68)	101 (13-149)	NR	NR
Thangarajah et al, ²⁴ 2017	8	IV	1	47 (21-73)	57 (24-123)	NR	NR
Thangarajah and Lambert, ²⁵ 2017	7	IV*	NR	48 (20-80)	98 (27-197)	NR	NR
Wagner et al, ²⁶ 2018	29	IV	10	42 (17-69)	144 (24-264)	NR	NR
Total	343 (316)	NA	213	NA	NA	NA	NA
Average	NA	NA	67.40%	38.4	65.55	53% [†]	3.98 (1.5-8)

LOE, level of evidence; NR, not reported; NA, not applicable.

All data were obtained during data extraction from the articles. The totals were summated and averaged to yield the presented values. Data for age, follow-up, and time to fusion are presented as mean (range).

* LOE based on Oxford Centre for Evidence-Based Medicine guidelines.

† Percentage of patients from listed studies who underwent prior surgery.

infections (n = 16, 18.6%) were the most commonly reported complications. However, of the entire cohort of patients (N = 316), 5.7% had humeral fractures and 5.0% had infections.

Discussion

This review demonstrates that shoulder arthrodesis can improve pain and function in patients with end-stage glenohumeral disease as a salvage treatment.²⁶ However, it is associated with a high complication rate that may make it less desirable than alternative treatments, which limits its utility. This is particularly important as patients who are candidates for shoulder arthrodesis tend to have complicated surgical histories with multiple prior operations and anatomic variations as a result.⁶

Arthrodesis is no longer considered a first-line treatment for glenohumeral arthritis and is rarely used as such.¹ However, it remains a viable and/or salvage option for certain disorders such as brachial and other neurologic plexopathies, severe and recalcitrant multidirectional glenohumeral instability with or without arthritis, and other rare conditions. Given the morbidity and relatively high rate of

complications, patients need to be examined and counseled thoroughly before proceeding with surgery. Lack of ipsilateral hand function is a relative contraindication to this procedure, as patients are usually only able to reach the hand of the affected limb to the midline of the front of the body. In addition, conservative management should still be considered before arthrodesis.¹ Fortunately, advances in arthroplasty including reverse shoulder replacement, which can be used in chronically unstable and other pathologic shoulders, have decreased the need to commonly perform fusions.

For patients who meet the criteria and elect to undergo shoulder arthrodesis, counseling about the significant risks including infection, fracture, failure to fuse, and other complications is required. Safran and Iannotti²¹ noted that the frequency of complications was directly correlated to the quality and quantity of bone available. Moreover, Rühmann et al²⁰ reported that 42% of patients had a fair assessment of their shoulder after arthrodesis whereas 7% had a poor assessment and 2% had a bad assessment after their procedure.

In patients who do not experience complications, functional improvements can be noted after glenohumeral fusions. The SSV was the most commonly reported functional outcome, and our findings showed that

Table II ROM after shoulder arthrodesis

	Active rotation, °	ROM, °			
		Abduction	Forward flexion	External rotation	Internal rotation
Authors					
Atlan et al, ² 2012	48 (15-85)	59 (20-110)	NR	NR	NR
Belkheyar et al, ³ 2019	NR	67 (50-85)	78 (60-95)	21 (0-40)	77
Chammas et al, ⁴ 2004 [*]	NR	61	Cohort A: 61 Cohort B: 62	Cohort A: 7 Cohort B: −9	Cohort A: 42 Cohort B: 45
Chib et al, ⁵ 2021	NR	60 (40-80)	65 (45-90)	NR	NR
Diaz et al, ⁸ 2003	NR	25 (15-40)	22 (20-30)	NR	35 (20-45)
Dimmen and Madsen, ⁹ 2007	NR	59 (20-80)	72 (20-90)	5 (−30 to 50)	23 (−20 to 40)
Lenoir et al, ¹² 2017	NR	59 (40-80)	80 (60-90)	NR	NR
Nagy et al, ¹⁷ 2004	NR	65	72	0	45
Rühmann et al, ²⁰ 2005	NR	56	60	NR	59
Sousa et al, ²³ 2011	NR	44	59	NR	NR
Average	48	57	64	3	48

ROM, range of motion; NR, not reported.

Internal rotation was measured via forearm to sagittal plane in the reported studies. Data are presented as mean (range) unless otherwise indicated.

* The study had 2 different cohorts (A and B).

Table III Time to fusion and fusion rate after shoulder arthrodesis

	Time to fusion, mean (range), mo	Fusion rate, %
Authors		
Belkheyar et al, ³ 2019	4.5 (3-6)	76
Chib et al, ⁵ 2021	5.4 (4.3-7.1)	100
Diaz et al, ⁸ 2003	3.5 (2.5-5)	100
Lenoir et al, ¹² 2017	3 (2-8)	100
Rühmann et al, ²⁰ 2005	3.5 (1.5-7)	83
Average	3.9	88.7

shoulder arthrodesis improved activities of daily living in all studies in which the SSV was used.^{19,24-26} The current review found that the SSV doubled after arthrodesis. This increase in the SSV after arthrodesis is in line with the hypothesis that shoulder arthrodesis still serves as a viable option to increase shoulder function and patient outcomes in patients with end-stage glenohumeral injuries. The results of this study were consistent with the findings of other studies showing the effectiveness of shoulder arthrodesis in regaining daily function of the shoulder.²¹

ROM, mostly provided through the scapula post-operatively, is an important outcome in arthrodesis patients, with a goal of providing a painless limb that allows increased ability to perform activities of daily living.²¹ Although there were limited data regarding the pre-arthrodesis ROM, the assumption is that this would be

severely limited. As pre-arthrodesis patients display limitations in ROM, there is a growing consensus that after arthrodesis, patients should be able to complete hygiene activities in the front of the body; however, moving the affected hand to the back is severely limited, and most patients can only reach the back pocket.²¹ In a study by Miller et al,¹⁵ abduction, forward flexion, and internal rotation were reported to average 42°, 24°, and 26° at an average follow-up of 41.3 months. Although their study was an evaluation of arthrodesis after flail shoulder in children, they reported that all patients were happy with their ROM and functional outcomes. With the data in this study providing higher ROM numbers, there may be an argument that the ROM data reported in this study may be linked to improved satisfaction, but this would need to be further evaluated.

Another factor to consider with the use of shoulder arthrodesis is the age of the patients. In this review, the average age of patients was 38 years and the late-stage ramifications remain to be elucidated. More review on the topic of arthrodesis in younger patients is warranted as the study by Miller et al¹⁵ was the only study identified evaluating a young patient cohort, with an average cohort age of 14.7 years. Another topic of interest is arthroscopically assisted fusions. Two different studies were found that compared outcomes between open arthrodesis and arthroscopically assisted arthrodesis; a full review of arthroscopic arthrodesis may shed more light on its utility.^{18,19} Finally, it is unknown whether cases of arthroscopically assisted fusion may perform differently than open cases especially regarding infection and fusion rates.

Table IV Functional outcomes associated with shoulder arthrodesis

	N	VAS score		SSV		ASES score	OSS	
		Preop	Postop	Preop	Postop		Preop	Postop
Authors								
Chammas et al, ⁴ 2004	27	NR	NR	NR	NR	NR	NR	NR
Diaz et al, ⁸ 2003	8	8.5	3.1	NR	NR	NR	NR	NR
Dimmen and Madsen, ⁹ 2007	18	NR	3 (1-8)	NR	NR	59 (15-95)	NR	32 (15-49)
Lenoir et al, ¹² 2017	8	NR	NR	NR	NR	69	NR	NR
Nagy et al, ¹⁷ 2004	20	NR	NR	NR	NR	NR	NR	NR
Puskas et al, ¹⁹ 2017	34	NR	NR	19	41	NR	NR	NR
Rühmann et al, ²⁰ 2005	43	NR	NR	NR	NR	NR	NR	NR
Scalise and Iannotti, ²² 2008	7	NR	NR	NR	NR	NR	NR	NR
Sousa et al, ²³ 2011	13	NR	NR	NR	NR	NR	NR	NR
Thangarajah et al, ²⁴ 2017	8	NR	NR	22 (0-50)	73 (50-100)	NR	8 (0-19)	32 (16-41)
Thangarajah and Lambert, ²⁵ 2017	7	NR	NR	7 (0-15)	45 (15-100)	NR	11 (4-16)	27 (16-40)
Wagner et al, ²⁶ 2018	23	NR	NR	NR	35	NR	NR	NR
Average	NA	8.5	3	18.8	43.9	62.1	9.4	30.9

VAS, visual analog scale; SSV, Subjective Shoulder Value; ASES, American Shoulder and Elbow Surgeons; OSS, Oxford Shoulder Score; NR, not reported; NA, not applicable.

Data are presented as mean (range) unless otherwise indicated.

Table V Further surgical procedures and complications after arthrodesis

	Rate of further surgery, %	Rate of complications reported, %
Authors		
Atlan et al, ² 2012	24.1	14.8
Chammas et al, ⁴ 2004	7.4	22.2
Chib et al, ⁵ 2021	NR	33.3
Diaz et al, ⁸ 2003	87.5	88
Dimmen and Madsen, ⁹ 2007	16.7	66.7
Lenoir et al, ¹² 2017	NR	25.0
Puskas et al, ¹⁹ 2017	41.2	20.6
Rühmann et al, ²⁰ 2005	37.2	32.6
Scalise and Iannotti, ²² 2008	57.1	57
Sousa et al, ²³ 2011	NR	23.1
Wagner et al, ²⁶ 2018	48.3	55.2
Average	38.2	33.6

NR, not reported.

Limitations

This study had limitations and sources of potential bias. This includes the limitations of the original works used for this study. The articles were all retrospective and uncontrolled, and this can lead to selection bias. There are also variations among surgical techniques for shoulder arthrodesis and patients' ages, which may affect the outcomes. Other limitations in this study include a small patient pool (N = 316). Moreover, regarding study functional outcomes, there were

fewer patients in the post-arthrodesis category than in the pre-arthrodesis category for many studies, meaning that for unknown reasons, functional data could not be obtained in certain pre-arthrodesis patients; this is a valid concern as it may bias the percent change values.

Conclusion

Shoulder arthrodesis provides improvement in functional outcomes for end-stage glenohumeral injuries; however, it is also associated with high rates of complications. Further research is needed regarding appropriateness criteria, as well as to determine whether contemporary techniques may be associated with lower rates of complications for this relatively rare procedure.

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